MASTER'S THESIS
(PHYSICS)

# Evolution of Open Quantum Systems under Continuous Measurement of their Environment

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*« La gloire de Dieu, c'est de cacher les choses; La gloire des rois, c'est de sonder les choses. »*
— **Proverbes 25:2**

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# CONTEXT AND MOTIVATIONS

This document is the report of an internship carried out as part of a Master's degree in fundamental and numerical physics (CompuPhys).
My internship has been conducted within the team of Theorical Physics — *équipe de Physique Théorique* ΦTh — of the *UTINAM Institute* in the building of the *Observatory of Besançon*.
It aims to study from an academical perspective at first, the subject of continuous measurement considering open quantum systems. The main objective was to explore a subject almost-unknown in details to either my supervisor or myself. We wanted to understand the possibilities and how to handle this domain properly. This work could then lead to other searches linked to continuous measurements.

By familiarizing ourselves with the topic, we decided to adopt this perticular frame of report. The structure of this document is meant to help any unfamiliar reader (about continuous measurement), to discover and explore its fundamentals. We wanted to make it as illustrative and pedagogical as possible, thus, simple examples will be used to graphically depict mathematical and physical concepts.

# Introduction

"Quantum mechanics is the description of the behavior of matter and light in all details and, in particular, of the happenings on an atomic scale" said *R.Feynman* in one of his lectures [1]. In this framework, position is no longer a single well-defined value prior to measurement, instead, position is described by a probability density all across space. Consequently, the notion of a trajectory as a continuous succession of points in spacetime is lost in quantum dynamics. The definition of trajectory will therefore be addressed, but one question remains paramount when mentioning systems: how can we even define a quantum system properly?

In the ideal case, systems are treated as closed, *i.e.*, isolated from any external environment and interacting with nothing at all. This modelling allows quantum mechanics to describe dynamics of systems with unitary and deterministic terms (the state at time $t$ is determined solely by knowing the state at time $t_0$) using the Schrödinger equation. [2] However in the physical world, systems do "feel" their surroundings, *i.e.*, they interact with their environment. This interaction should be taken into account to obtain a more complete and physically relevant representation. In doing so, we describe the so-called *Open Quantum System* (OQS) theory [3].

Other useful theories have been developed through the last decades, allowing quantum mechanics' description to be more and more complete. One of them being the Quantum Measurement Theory (QMT) [4], where the concept of measurement is fully described. Measuring a system has a tremendous impact on the its quantum properties. Furthermore, since any experimental probe inevitably interacts with the system, it is crucial to figure out the nature of this interaction. Measuring a system without impacting its quantum properties leads to the concept of weak measurements, that we will develop further below.

By studying QMT, one can thus notice that measuring a quantum system alters it. The measurement usually interacts so strongly with the system that it collapses into a state — an eigenstate of the observable — that lost interesting quantum properties. For example it can lose its entanglement[1]. Along the same lines, quantum systems can be in superposition of states, taking *Schrödinger's cat* thought experiment, this corresponds formally to a superposition of 'alive' and 'dead' states prior to measurement. Measuring the state of the cat will determine its fate, alive or dead, it is then "stuck" in the state measured and cannot be written as a superposition again. The idea of **weak measurements** is to **measure the environment** coupled with the system. Instead of altering the state, we obtain pieces of information on the system. The measurements can even be conducted continuously, maximizing the amount of data the observer can get. These are the global ideas behind **Quantum Continuous Measurements** (QCM) [5].

Practically, this work will be organized as follows: the first section covers an academical progression from basics in quantum mechanics up to the aforementioned theories. A second section will cover the photodetection and homodyne detection of open quantum systems. The qubit and harmonic oscillator will be used to illustrate our point. The final part is the study of a system of interest on which we apply quantum continuous measurement. (Si on a le temps.)

[1] $|\psi\rangle_{AB} = \frac{|00\rangle+|11\rangle}{\sqrt{2}} \xrightarrow[\text{in space A}]{\text{measuring 0}} |00\rangle_{AB} = |0\rangle_A \otimes |0\rangle_B$ a product state (thus non-entangled).

# I From closed systems to continuous measurements — in a nutshell

In this chapter we will briefly summarize the definitions needed for continuously monitored open quantum systems formalism.

## I.i) Generalities on quantum mechanics

### I.i.1⟩ States and time evolution

In quantum mechanics a physical system's state is described by a vector in a Hilbert space:

$$|\psi\rangle \in \mathcal{H}. \tag{1}$$

The Hamiltonian $\hat{H}$ is a specific hermitian[2] operator representing classically the total energy of a system; this relationship will be detailed in Section I.i.2⟩. Physically, the evolution of a system is governed by its energy, the *Schrödinger equation* states it explicitly:

$$i\hbar \frac{\mathrm{d}\,|\psi(t)\rangle}{\mathrm{d}t} = \hat{H}(t)\,|\psi(t)\rangle\,. \tag{2}$$

We can define a unitary[3] operator $\hat{U}$ such that,

$$|\psi(t)\rangle = \hat{U}(t,t_0)\,|\psi(t_0)\rangle\,, \tag{3}$$

solving formally the Schrödinger equation. By injecting the previous relation in the state equation, we get

$$\frac{\mathrm{d}\hat{U}}{\mathrm{d}t}(t,t_0) = -\frac{i}{\hbar}\hat{H}(t)\hat{U}(t,t_0). \tag{4}$$

In the general case, the Hamiltonian can depend on time, so we need the chronological time-ordering exponential [3] to solve Eq. (4). However, in the simplest case, the Hamiltonian is independent of time and the solution becomes

$$|\psi(t)\rangle = \underbrace{e^{-\frac{i\hat{H}(t-t_0)}{\hbar}}}_{\hat{U}(t,t_0)}\,|\psi(t_0)\rangle\,. \tag{5}$$

This solution of the Schrödinger equation is indeed **unitary** and we can use $\hat{U}^\dagger(t,t_0)$ on the left of both sides of Eq. (5) to obtain $|\psi(t_0)\rangle = \hat{U}^\dagger(t,t_0)\,|\psi(t)\rangle$, which proves that the dynamics of closed systems are **reversible**.

### I.i.2⟩ Observables

In quantum mechanics, any physical quantity is represented by an **observable** which is an operator fulfilling two properties. The eigenvectors of the observable form an orthonormal basis of the Hilbert space $\mathcal{H}$, moreover the corresponding eigenvalues are necessarily

[2] $\hat{H}^\dagger = \hat{H}$, where $^\dagger$ denotes the conjugate transpose, defining the hermitian adjoint.

[3] $\hat{U}\hat{U}^\dagger = \hat{U}^\dagger\hat{U} = \mathbb{I}$, where $\mathbb{I}$ is the identity.

real [2]. For finite dimension Hilbert space, hermitian operators[4] are observables. In the following, we will consider finite dimension and non-degenerate cases, b [2].

A natural operator whose eigenbasis is particularly interesting is the Hamiltonian, whose eigenvalues correspond to the allowed energy values of a system. Let us denote $I_{\mathcal{H}} = [\![1; \dim(\mathcal{H})]\!]$ throughout this document.

$$\forall n \in I_{\mathcal{H}}, \quad H\left|\phi_n\right\rangle = E_n\left|\phi_n\right\rangle \tag{6}$$

Let us call $\{\phi_n\}_{n\in I_{\mathcal{H}}}$ the family of orthonormal eigenvectors of $H$ which forms a basis of $\mathcal{H}$.Since it forms a basis of the Hilbert space, one can describe any state in this very basis:

$$\forall \left|\psi\right\rangle \in \mathcal{H}, \qquad \left|\psi\right\rangle = \sum_{n\in I_{\mathcal{H}}} \underbrace{c_n}_{\langle\phi_n|\psi\rangle} \left|\phi_n\right\rangle \tag{7}$$

This kind of description in a base is obviously valid for any observable, not only for the Hamiltonian. We can complete this part with the normalization of the state, for proper handling of scalar products, we need to set $|\psi| = \sqrt{\langle\psi|\psi\rangle} = 1$. The condition $\langle\psi|\psi\rangle = \sum_n |c_n|^2 = 1$ is thus equivalent.

### I.i.3⟩ Purity of states

Taking the qubit example, let $\alpha, \beta \in \mathbb{C}$ to write the qubit state as a superposition of excited/ground state as:

$$\left|\psi\right\rangle = \alpha\left|e\right\rangle + \beta\left|g\right\rangle \qquad \text{where, } |\alpha|^2 + |\beta|^2 = 1 \tag{8}$$

By getting ahead of QMT, we can introduce the probability to find a system in one of the possible eigenstates of an observable. Let us take the following example, what is the probability to find the energy $E_n$, defined by Eq. (6), from a system in $\left|e\right\rangle$ state? From a system in $\left|g\right\rangle$ state?

$$\mathbb{P}_e[E_n] = |\langle\phi_n|e\rangle|^2 \tag{9}$$

$$\mathbb{P}_g[E_n] = |\langle\phi_n|g\rangle|^2 \tag{10}$$

A physical state can be, by the concepts of quantum mechanics, a mixture of statistical probabilities [2]. Let us now consider back the superposition from Eq. (8), a physical mixture would be described by mixing states $\left|e\right\rangle, \left|g\right\rangle$ with weights $|\alpha|^2, |\beta|^2$:

$$\mathbb{P}_\psi[E_n] \stackrel{?}{=} |\alpha|^2\ \mathbb{P}_e[E_n] + |\beta|^2\ \mathbb{P}_g[E_n] \tag{11}$$

In fact, quantum mechanics gives the exact value of this probability:

$$\mathbb{P}_\psi[E_n] = |\langle\phi_n|\psi\rangle|^2 \tag{12}$$

$$= |\alpha|^2|\langle\phi_n|e\rangle|^2 + |\beta|^2|\langle\phi_n|g\rangle|^2 + 2\,\mathrm{Re}\left\{\alpha\beta^*\,\langle\phi_n|e\rangle\,\langle\phi_n|g\rangle^*\right\} \tag{13}$$

[4]Hermitian operators always form a basis in finite dimension, and, we can use the Gram-Schmidt process to find its orthonormal version.Moreover, being a self-adjoint operator implies having real eigenvalues.

The statistical mixture terms appear but also cross terms in the true probability of finding the superposition in one of the eigenstates. This notion of hidden correlations due to superposition has been highlighted by a new formalism, **density matrices**.

Basically, instead of a ket $|\psi\rangle$ we describe a system by a density matrix $\rho$ which, for **pure** systems is simply:

$$\rho_{\mathrm{p}} = |\psi\rangle\langle\psi| \tag{14}$$

The density matrix is thus an operator of Hilbert space elements[5] $\rho : \mathcal{H} \to \mathcal{H}$. For a **statistical mixture**, thus non-pure, the density matrix in a basis $\{\phi_n\}_{n\in I_{\mathcal{H}}}$ takes the form:

$$\rho_{\mathrm{sm}} = \sum_{n\in I_{\mathcal{H}}} \mathbb{P}_n \, |\phi_n\rangle\langle\phi_n| \tag{15}$$

The components of the matrix are precisely, on the diagonal, the probabilities to find the state of the system in one of the eigenstates of the observable. The off-diagonal components are called **coherences**, they represent interference effects between the eigenstates, they can appear when the system's state is a linear coherent superposition of these eigenstates [2].

The mathematical definition of a density matrix is in any case [2, 3]:

- $\rho$ is hermitian $\rho^\dagger = \rho$
- $\rho$ is semi-definite $\forall\, |\alpha\rangle \in \mathcal{H}, \quad \langle\alpha|\, \rho\, |\alpha\rangle \geq 0$
- $\rho$ respects $\mathrm{Tr}\,[\rho] = 1$

### I.i.4⟩ Postulates of quantum mechanics

Physicists developed a corpus of the main properties of quantum mechanics, here are summarized the so-called postulates of quantum mechanics [2]:

1. The state of a system is represented by a density matrix $\rho \in \mathcal{L}(\mathcal{H})$. For pure states we can describe them using the Dirac formalism $|\psi\rangle \in \mathcal{H}$, or using the pure matrix density formalism $\rho_p = |\psi\rangle\langle\psi|$.
2. Physical quantities $\mathcal{A}$ are associated with operators $\hat{A}$, called observables. In the context of finite dimension space, they are simply Hermitian operators.
3. The only possible outcomes of a measurement of $\mathcal{A}$ are the eigenvalues $a$ of $\hat{A}$, defined by $\hat{A}\,|a\rangle = a\,|a\rangle$.
4. When measuring $\mathcal{A}$ on a normalized state, we know the probability to get one of the possible eigenvalues: $\mathbb{P}(a_n)$ (which depends on the spectrum of the observable).
5. If the measurement of the physical quantity $\mathcal{A}$ on the system gives the result $a_n$, the state of the system immediately after the measurement collapses into the eigenstate $|\phi_n\rangle$.

[5] $\rho$ is also an operator of the dual, $\mathcal{H}^*$ because we find that $\langle\phi_j|\, \rho \propto \langle\phi_j|$.
We could even ague that $\rho \in \mathcal{L}(\mathcal{H})\ldots$

6. The time evolution of the wavefunction $|\psi(t)\rangle$ is governed by the *Schrödinger* equation:

$$\frac{\mathrm{d}\,|\psi(t)\rangle}{\mathrm{d}t} = -\frac{i}{\hbar} H(t)\,|\psi(t)\rangle$$

For density matrices, the evolution is driven by the *Liouville-von Neumann* equation[6]:

$$\frac{\mathrm{d}\rho}{\mathrm{d}t}(t) = -\frac{i}{\hbar}\,[H(t), \rho(t)] \equiv \mathcal{L}(t)\rho(t) \tag{16}$$

[6] $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$ is the commutator of $\hat{A}$ and $\hat{B}$

## I.ii) Open Quantum Systems (OQS)

In the previous part, we saw that one can fully describe a system, observables, states and their evolution. We recall that this description is ideal in the sense that the system is considered to be completely isolated from any interaction with its environment. In this section, we introduce the concepts of OQS which lead to a more physical description of quantum mechanics.

It is important to understand that the total system is a closed system, the total system space is mathematically $\mathcal{H}_T = \mathcal{H}_S \otimes \mathcal{H}_E$ . The total system $T$ thus follows a unitary time evolution, but, it is not the case for the physical system of interest $S$. Another of dynamics will arise for $S$ due to its coupling with $E$. this is why we call $S$ an open quantum system.

By following the formalism of reference [3], we need to jump in a different representation of the equations: the interaction picture. In the *Schrödinger* picture, states evolve with respect to time, observables can be constants or depend explicitly on time. Instead, in the *Heisenberg* picture it is the opposite, observables carry the whole dynamics and states are constants (see Eq. A. 2). The interaction picture is a mix of both, where states and observables are functions of time.

Interaction picture begins with the perturbation theory, where we split the Hamiltonian in two parts, the free Hamiltonian $H_0$ and the interaction Hamiltonian $H_I$

$$H_T = \overbrace{H_S \otimes \mathbb{I}_E + \mathbb{I}_S \otimes H_E}^{H_0} + H_I \tag{17}$$

$$U(t, t_0) = U_0(t, t_0) U_I(t, t_0) \tag{18}$$

We define the interaction Hamiltonian by,

$$E_n(t) = U_0^\dagger(t) E_n U_0(t) \tag{19}$$

$$H_I(t) = \sum_{n \in I_{\mathcal{H}}} \overbrace{|n\rangle\langle n|}^{A_n} \otimes E_n(t) \tag{20}$$

We can then find the useful following relations:

$$\tilde{\rho}(t) = U_I(t) \rho_S U_I^\dagger(t) = U_0^\dagger(t) \rho_S(t) U_0(t) \tag{21}$$

$$\tilde{O}(t) = U_0^\dagger(t) O_S(t) U_0(t) \tag{22}$$

One can then find the equation that governs the dynamics of $\rho_S$[7] in the *Schrödinger picture* by making progress in the *interaction picture*. We call this equation the *Redfield Master Equation* but its expression is analytically formal. To go further, one needs to make important assumptions known as the *Born-Markov* approximations and the *secular* approximation.

[7] Here and in the following, $\rho_S$ is the state of the system obtained by tracing out the environnement from the total state: $\rho_S = \mathrm{Tr}_E[\rho]$

♦ The *Born* approximation consists in saying that the environment acts as a thermodynamical reservoir, *i.e.*, the environment is so big compared to the system that the interaction does not alter its state:

$$\tilde{\rho}_T(t) = \tilde{\rho}_S(t) \otimes \rho_E \tag{23}$$

It describes a so-called "weak-coupling" between the system and the environment.

♦ The *Markov* approximation assures that the entire system has no memory. That is to say, the environment is so rapidly agitating that it dissipates information of the system instantaneously. This implies that the environment comes back to equilibrium instantaneously after each interaction. Physically, the timescale of the reservoir correlation function decay $\tau_E$ is smaller than the timescale $\tau_R$ on which the system varies "appreciably".

These two last approximations lead to the *Markovian quantum master equation.*

♦ The *secular* approximation allows the possibility to neglect rapidly oscillating terms in the quantum *Markov* master equation.

## I.iii) Quantum Measurement Theory (QMT)

In the QMT [6, 7], measuring consists in applying a projector to a state, so, we call this process a *projective measurement*. The set of projective measurements is called a Projection-Valued Measure (PVM) [8]. For the sake of simplicity, we work — as until now — with non-degenerate spectra for the observables. Since an observable has a set of eigenvectors $\{|a_n\rangle\}_{n\in I_{\mathcal{H}}}$, we can build a projector corresponding to one of them:

$$P_{a_n} = |a_n\rangle\langle a_n| \,. \tag{24}$$

This definition implies four important properties:

$$P_{a_n}^\dagger = P_{a_n}, \qquad P_{a_n}^2 = P_{a_n}, \qquad P_{a_n}P_{a_m} = P_{a_n}\delta_{nm}, \qquad \sum_n P_{a_n} = \mathbb{I}.$$

To represent the quantum mechanics definition of a measurement, the projective measurement needs to fulfill some requirements. The first requirement is to generate the probability of observing the result of a measurement on a system. Thus, the probability to observe $a_n$ is defined by

$$\mathbb{P}_{|\psi\rangle}[a_n] = \langle\psi|P_{a_n}|\psi\rangle = |\langle\psi|a_n\rangle|^2, \tag{25}$$

$$\mathbb{P}_{\rho}[a_n] = \mathrm{Tr}\left[\rho P_{a_n}\right], \tag{26}$$

where the subscripts $|\psi\rangle$ and $\rho$ refer respectively to the representation of pure states or a general density matrix. Moreover, the theory of quantum mechanics stipulates that states collapse on the eigenstate corresponding to the result after measurement, thus, their expression after collapse is

$$|\tilde{\psi_a}\rangle = P_a\,|\psi\rangle\,, \tag{27}$$

$$\tilde{\rho_a} = P_a\rho P_a. \tag{28}$$

Here the $\tilde{\bullet}$ symbol means that the state is not normalized, which will be an important notation for later, see Section **??**. By normalizing we find the proper expression of a state after a measurement with result $a_n$:

$$|\psi_a\rangle = \frac{P_a\,|\psi\rangle}{\sqrt{\langle\psi|P_a|\psi\rangle}}, \tag{29}$$

$$\rho_a = \frac{P_a\rho P_a}{\mathrm{Tr}\left[\rho P_a\right]}. \tag{30}$$

The obtained state is said **conditioned by the measurement** result $a_n$, hence the importance of the index $a_n$ on states obtained after measurement. We point out that the normalization factor is exactly the probability to observe the result $a_n$ for the density matrix (or the square root of the probability, when using ket formalism). In the end, projective measurement fulfill the properties of quantum mechanics by construction because they generate probabilities (respecting the fact they need to be positive and their sum is 1), and can also be used to write the expression of collapsed states after measurement.

However, the definition of PVM has been found too restrictive to represent every kind of measurement allowed by quantum mechanics [9, 10]. One can define a more general kind of measurement, that fulfills quantum mechanics needs while relaxing some properties.

In particular, the orthogonality of measure elements is not required in order to generate probabilities. Therefore, one can build the so-called Positive Operator-Valued Measure (POVM), implicitly defined by the probability to observe the result $a_n$:

$$\mathbb{P}_\psi[a_n] = \langle\psi|\Pi_{a_n}|\psi\rangle\,, \tag{31}$$

$$\mathbb{P}_\rho[a_n] = \mathrm{Tr}\left[\rho\Pi_{a_n}\right], \tag{32}$$

where the operator $\Pi_{a_n}$ is called the *effect* [7] for the result $a$. Moreover, the set of all effects $\{\Pi_{a_n}\}_{n\in I_\mathcal{H}}$ forms the POVM. As stated, the only properties we force the POVM to respect are:

$$\text{positive semi-definite: } \forall\,|\phi\rangle \in \mathcal{H},\ \langle\phi|\Pi_{a_n}|\phi\rangle \geq 0,\ \text{and,} \qquad \sum_{n\in I_\mathcal{H}} \Pi_{a_n} = \mathbb{I}.$$

These properties imply that POVM generate proper probabilities. Furthermore, the (normalized) states obtained after finding the result $a_n$ are now,

$$|\psi_{a_n}\rangle = \frac{K_{a_n}\,|\psi\rangle}{\sqrt{\langle\psi|\Pi_{a_n}|\psi\rangle}}, \tag{33}$$

$$\rho_{a_n} = \frac{K_{a_n}\rho K_{a_n}^\dagger}{\mathrm{Tr}\left[\rho\Pi_{a_n}\right]}, \tag{34}$$

where we introduced the operator $K_{a_n}$ called a *Kraus operator*; they respect the relation $\Pi_{a_n} = K_{a_n}^\dagger K_{a_n}$.

Also, POVM, just like PVM, allow one to describe the process of quantum measurement while *ignoring the results*. As in the thought experiment **??** for statistical mixture of states, we can describe statistically the state without knowing the measurement results:

$$\rho = \sum_{n\in I_\mathcal{H}} \mathbb{P}[a_n]\rho_{a_n} = \sum_{n\in I_\mathcal{H}} K_{a_n}\rho K_{a_n}^\dagger. \tag{35}$$

The corresponding relation for PVM is obtained by replacing both $K_{a_n}$ and $K_{a_n}^\dagger$ by $P_{a_n}$.

Finally, one can observe that POVM generalize PVM in the sense that projectors are special cases of POVM elements (adding self-adjoint property and orthogonality to POVM gives a PVM). Furthermore, PVMs make states collapse on eigenstates that are necessarily orthogonal to each other, which is not necessarily the case for POVMs. A brief example of POVM forcing a state to collapse onto "non-orthogonal states" is when considering initially the state $|\psi\rangle = \frac{|g\rangle+|e\rangle}{\sqrt{2}} \equiv |+\rangle$. If we choose the Kraus operators $K_0 = |g\rangle\langle g|$ and $K_1 = |+\rangle\langle e|$, which fulfill $K_0^\dagger K_0 + K_1^\dagger K_1 = \mathbb{I}$, and, that form the POVM $\{|g\rangle\langle g|\,, |e\rangle\langle e|\}$. Then, when the result of the measurement is 0, we find that the state collapses to $|\psi_0\rangle = |g\rangle$, and when finding 1 it collapses to $|\psi_1\rangle = |+\rangle$. However, $\langle\psi_0|\psi_1\rangle = \frac{1}{\sqrt{2}} \neq 0$, thus the states on which the initial state can collapse are not orthogonal to each other.

# II Continuously measured quantum systems – obtaining a Stochastic Master Equation (SME)

# A Representations of quantum mechanics

**Complement of quantum mechanics:**
An average is defined in quantum mechanics by

$$\begin{aligned}\langle O\rangle_\psi &= \langle\psi|O|\psi\rangle\,,\\ \langle O\rangle_\rho &= \mathrm{Tr}\left[\rho O\right].\end{aligned}$$

Both definitions are strictly equivalent, to show it, let us pick a pure state $\rho = |\psi\rangle\langle\psi|$ and the base of an Hamiltonian $\{\phi_n\}_{n\in I_\mathcal{H}}$:

$$\begin{aligned}\langle O\rangle_\rho &= \mathrm{Tr}\left[\rho O\right]\\ &= \mathrm{Tr}\left[|\psi\rangle\langle\psi|\,O\right]\\ &= \mathrm{Tr}\left[\,\langle\psi|O|\psi\rangle\right].\end{aligned}$$

However, trace of a scalar is equal to this scalar, thus,

$$\langle O\rangle_\rho = \langle\psi|O|\psi\rangle\,.$$

## A.a) Schrödinger and Heisenberg pictures

In this appendix we will denote the pictures with their acronyms, and, we lighten the notation of the total state $\rho_T \equiv \rho$ here. This could be misleading: $S$ in this appendix stands for *"in the Schrödinger picture"*, whereas, $S$ in the main chapters means *"state of the system S"*.

In the Schrödinger picture we saw that $|\psi(t)\rangle = U(t)\,|\psi\rangle$, where $U(t) = e^{-\frac{i}{\hbar}Ht}$ in the case of a time-independent Hamiltonian with $t_0 = 0$. Nevertheless, the following results can be derived in the general case of a time dependent Hamiltonian, [3]. Considering the average value of an observable $A_S(t)$ in the Schrödinger picture, we find

$$\begin{aligned}\langle A_S(t)\rangle &= \langle\psi(t)|A_S|\psi(t)\rangle\\ &= \langle\psi|\underbrace{U^\dagger(t)A_S(t)U(t)}_{A_H(t)}|\psi\rangle\,.\end{aligned} \tag{A. 1}$$

This calculation naturally reveals the Heisenberg representation of observables and states. Indeed, we have that $\forall t,\ |\psi(t)\rangle_H = |\psi(t_0)\rangle_S \equiv |\psi\rangle_H$ which links both pictures at the initial time. Furthermore, we can link both representation by calculating the evolution of an observable in the Heisenberg picture:

$$\boxed{\frac{\mathrm{d}A_H}{\mathrm{d}t} = \frac{i}{\hbar}\left[H_H, A_H\right] + \frac{\partial A_S}{\partial t}}. \tag{A. 2}$$

This link can even lead to the famous *Ehrenfest theorem* for quantum mechanics:

$$\boxed{\frac{\mathrm{d}\,\langle A\rangle}{\mathrm{d}t} = \frac{i}{\hbar}\left\langle\left[H(t), A(t)\right]\right\rangle + \left\langle\frac{\partial A}{\partial t}\right\rangle}. \tag{A. 3}$$

We observe that if the observable doesn't depend explicitly on time (thus constant), then the dynamics of the operator is fully describe by the Heisenberg representation.

## A.b) Interaction picture

As mentioned in I.ii), we use perturbation theory to describe the interaction between the system $S$ and the environment $E$,

$$H_T = \overbrace{H_S \otimes \mathbb{I}_E + \mathbb{I}_S \otimes H_E}^{H_0} + H_I, \tag{17}$$

$$U(t, t_0) = U_0(t, t_0) U_I(t, t_0). \tag{18}$$

Because $|\psi(t)\rangle_S = U(t)\,|\psi\rangle_S$, we can write that

$$\rho_S(t) = U(t)\rho_S U^\dagger(t). \tag{A. 4}$$

If we now try to calculate the average of an operator (in the total space $\mathcal{H}_T$) using the Schrödinger picture, we can find a new interesting relation by injecting Eq. (A. 4,18) in the average:

$$\begin{aligned}\langle O_S(t)\rangle &= \mathrm{Tr}\,[\rho_S(t) O_S(t)] \\ &= \mathrm{Tr}\,[\underbrace{U_I(t)\rho_S U_I^\dagger(t)}_{\tilde{\rho}(t)}\ \underbrace{U_0^\dagger(t) O_S(t) U_0(t)}_{\tilde{O}(t)}].\end{aligned}$$

We can thus define in the interaction picture:

$$\tilde{\rho}(t) = U_I(t)\rho_S U_I^\dagger(t), \tag{21}$$

$$\tilde{O}(t) = U_0^\dagger(t) O_S(t) U_0(t). \tag{22}$$

We can now reverse Eq. (A. 4) and then use Eq. (18):

$$\rho_S = U^\dagger(t)\rho_S(t)U(t) = U_I^\dagger(t)U_0^\dagger(t)\rho_S(t)U_0(t)U_I(t),$$

and, by injecting this into Eq. (21), we get

$$\begin{aligned}\tilde{\rho}(t) &= U_I(t)U_I^\dagger(t)U_0^\dagger(t)\rho_S(t)U_0(t)U_I(t)U_I^\dagger(t) \\ &= U_0^\dagger(t)\rho_S(t)U_0(t).\end{aligned} \tag{21}$$

So that we find the results used in the main part of this work

$$\boxed{\tilde{\rho}(t) = U_I(t)\rho_S U_I^\dagger(t) = U_0^\dagger(t)\rho_S(t)U_0(t)}. \tag{A. 5}$$

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